

Project Design Phase

Problem – Solution Fit Template

Date	29 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem–Solution Fit for "Plugging into the Future: An Exploration of Electricity Consumption Patterns" means identifying the challenges faced by energy planners, utility providers, researchers, and policymakers in understanding electricity consumption trends, and providing an interactive data visualization solution that effectively addresses these challenges. The project transforms complex electricity consumption data into meaningful insights through interactive dashboards, enabling users to analyze patterns, compare regions, and make informed decisions.

Purpose:

- ❑ **Analyze** complex electricity consumption data in a simple and interactive manner that meets the needs of energy planners, researchers, and decision-makers.
- ❑ **Improve** decision-making by identifying consumption trends, seasonal variations, and regional differences through visual analytics.
- ❑ **Provide** clear, interactive dashboards that enable users to explore electricity consumption data using filters, charts, and comparative visualizations.
- ❑ **Support** energy planning and policy development by identifying high and low electricity-consuming regions and understanding changes caused by factors such as the COVID-19 pandemic

Template:

Problem – Solution Fit Template:

Project: "Plugging into the future: An exploration of electricity consumption patterns."

Define the Problem (P)	1. CUSTOMER SEGMENT(S) CS Who is your customer? eg. Households, utility companies, energy analysts, policymakers, sustainability researchers.	5. CUSTOMER CONSTRAINTS CC What constraints or challenges do your customers face in getting insights into electricity consumption? eg. Limited real-time data, complex patterns, lack of awareness, high energy costs, difficulty in predicting usage.	6. AVAILABLE SOLUTIONS AS What solutions are available in the market today to solve these problems? eg. Smart meters, energy dashboards, utility apps, spreadsheets, basic analytics tools.	Explore Existing Alternatives (A)
	2. JOBS-TO-BE-DONE / PROBLEMS J&P What jobs do your customers hire your product to do? What problems are they trying to solve? eg. Monitor electricity usage, reduce wastage, identify high consumption areas, forecast demand, optimize energy use.	3. PROBLEM ROOT CAUSE RC What is the real reason behind these problems? eg. Data is scattered, lack of real-time insights, no predictive analysis, complex data hard to interpret, limited visualization.	7. BEHAVIOUR BE How do your customers behave when trying to solve these problems with existing solutions? eg. Manually tracking usage, exporting data, creating Excel reports, spending hours analyzing trends.	Dig Deeper into Customer Behaviour (B)
Identify the Right Problem (I)	4. TRIGGERS TR What triggers your customers to start looking for a solution? eg. High electricity bills, seasonal changes, energy audit requirements, policy changes, sustainability goals.	8. YOUR SOLUTION SL How does your solution solve the customer's problem? eg. An interactive dashboard that visualizes electricity consumption patterns, identifies trends, provides real-time insights and forecasts to help users make data-driven decisions.	9. CHANNELS of BEHAVIOUR CH How do customers access your solution? eg. Web application, mobile app, utility portals, reports, API integrations, cloud platforms.	Measure & Validate the Impact (M)
	4. EMOTIONS: BEFORE / AFTER EM How do customers feel before and after solving the problem? eg. Before: Frustrated, confused, concerned After: Informed, in control, confident, satisfied.		10. METRICS How will you measure that your solution is working? eg. Dashboard usage, insights generated, energy savings, accuracy of forecasts, user satisfaction.	

